ECON 3201
Econometrics for Economics and Finance

Assignment 2
Due November 5, 2026

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Instructions

Complete your midterm exam using R and type your answers in LaTeX. Answer all questions. Please complete your assignment using Quarto. Upload both your Quarto file and pdf file to your website under new blog post.

Questions

Question 1

X and Y are discrete random variables with the following joint distribution. Use the data frame `d` to answer your questions:

```
library(kableExtra)
d <- data.frame('\\hspace{1pt}' = c(rep('Value of X',3)),
                '\\hspace{1pt}' = c(1,5,8),
                '20' = c(0.07,0.32,0.05),
                '30' = c(0.10,0.05,0.15),
                '40' = c(0.04,0.03,0.19),
                check.names = FALSE)


knitr::kable(d,
             format = "latex",
             booktabs = T,
             escape = FALSE,
             align = c("l","c","c","c","c"),
             longtable= T )%>%
  kable_styling(position = "center")%>%
```

```
  add_header_above(c(" " = 1, " " = 1, "Value of Y" = 3))%>%
  collapse_rows(columns = 1:2, valign = "top")
```

Table 1: Joint Distribution of X and Y

		Value of Y		
		20	30	40
Value of X	1	0.07	0.10	0.04
	5	0.32	0.05	0.03
	8	0.05	0.15	0.19

a) Calculate the marginal probability distribution of Y. [2 Marks]

$$\Pr(Y = 20) = 0.07 + 0.32 + 0.05 = 0.44$$
$$\Pr(Y = 30) = 0.10 + 0.05 + 0.15 = 0.30$$
$$\Pr(Y = 40) = 0.04 + 0.03 + 0.19 = 0.26$$

```
joint <- matrix(c(0.07, 0.32, 0.05, 0.10, 0.05, 0.15, 0.04, 0.03, 0.19),
                nrow = 3,
                dimnames = list(X = c("1", "5", "8"),
                                Y = c("20", "30", "40")))

py <- colSums(joint)
py
```

```
  20   30   40
0.44 0.30 0.26
```

b) Calculate the mean of Y. [3 Marks]

$$E(Y) = 20(0.44) + 30(0.30) + 40(0.26) = 28.2$$

```
EY <- sum(c(20, 30, 40) * py)
EY
```

```
[1] 28.2
```

c) Calculate the variance of Y. [3 Marks]

$$\begin{aligned} Var(Y) &= \sum [y - E(Y)]^2 * \Pr(Y = y) \\ &= (20 - 28.2)^2(0.44) + (30 - 28.2)^2(0.30) + (40 - 28.2)^2(0.26) \\ &= 66.76 \end{aligned}$$

```
variance_y <- sum((c(20, 30, 40) - EY)^2 * py)
variance_y
```

```
[1] 66.76
```

Question 2

Using R commands, compute the following probabilities:

a) If Y is distributed $N(3, 9)$, find $Pr(Y > 1)$. [2 Marks]

$$\begin{aligned} \Pr(Y > 1) &= \Pr\left(Z > \frac{1-3}{\sqrt{9}}\right) \\ &= \Pr\left(Z > -\frac{2}{3}\right) \\ &= 1 - \Phi\left(-\frac{2}{3}\right) \\ &= 0.7475 \end{aligned}$$

```
1 - pnorm(1, mean = 3, sd = sqrt(9))
```

```
[1] 0.7475075
```

b) If Y is distributed $N(5, 2)$, find $Pr(6 \leq Y \leq 8)$. [3 Marks]

$$\begin{aligned} \Pr(6 \leq Y \leq 8) &= \Pr\left(\frac{6-5}{\sqrt{2}} \leq Z \leq \frac{8-5}{\sqrt{2}}\right) \\ &= \Phi\left(\frac{3}{\sqrt{2}}\right) - \Phi\left(\frac{1}{\sqrt{2}}\right) \\ &= 0.2228 \end{aligned}$$

```r
pnorm(8, mean = 5, sd = sqrt(2)) - pnorm(6, mean = 5, sd = sqrt(2))
```

```
[1] 0.2228026
```

Question 3

The random variable Y be a random variable with mean μ and variance σ^2. Type your answers.

a) Let Z denote the standardized version of Y. What is the functional form of Z? [1 Mark]

$$Z = \frac{Y - \mu}{\sigma}$$

b) Show that the expected value of of Z, $E(Z)$, in part (a) is 0. [3 Marks]

$$\begin{aligned} E(Z) &= E\left(\frac{Y-\mu}{\sigma}\right) \\ &= \frac{1}{\sigma}[E(Y) - \mu] \\ &= \frac{1}{\sigma}(\mu - \mu) \\ &= \frac{1}{\sigma}(0) \\ &= 0 \end{aligned}$$

c) Show that the variance of Z, $E\left[(Z - E(Z))^2\right]$, in part (a) is 1. [3 Marks]

$$\begin{aligned} E[(Z - E(Z))^2] &= E[(Z-0)^2] \\ &= E(Z)^2 \\ &= E[\left(\frac{Y-\mu}{\sigma}\right)^2] \\ &= \frac{1}{\sigma^2}E[(Y-\mu)^2] \\ &= \frac{\sigma^2}{\sigma^2} \\ &= 1 \end{aligned}$$

Question 4

Table 2 displays the joint probabilities of covid-19 vaccination status and hospitalization with covid-19.

```
library(kableExtra)
d <- data.frame('\\hspace{1pt}' = c('Hospitalized',''),
                '\\hspace{1pt}' = c('Yes','No'),
                'Yes' = c(0.007,0.666),
                'No' = c(0.075,0.252),
                check.names = FALSE)


knitr::kable(d,
             format = "latex",
             booktabs = T,
             escape = FALSE,
             align = c("l","c","c","c"),
             longtable= T)%>%
  kable_styling(position = "center")%>%
  add_header_above(c(" " = 1, " " = 1, "Vaccinated" = 2))
```

Table 2: Joint Distribution of Covid-19 Vaccination Status and Hospitalizations Due to Covid-19

		Vaccinated	
		Yes	No
Hospitalized	Yes	0.007	0.075
	No	0.666	0.252

a) Compute the probability that an individual is vaccinated conditional on being hospitalized with covid-19, i.e. $Pr(\text{Vaccinated} = \text{Yes}|\text{Hospitalized} = \text{Yes})$. [2 Marks]

$$\begin{aligned}
\Pr(\text{Vaccinated=Yes}|\text{Hospitalized=Yes}) &= \frac{\Pr(\text{Vaccinated=Yes, Hospitalized=Yes})}{\Pr(\text{Vaccinated=Yes})} \\
&= \frac{0.007}{0.007 + 0.075} \\
&= \frac{0.007}{0.082} \\
&= 0.0854
\end{aligned}$$

```
joint <- matrix(c(0.007, 0.666, 0.075, 0.252), nrow = 2,
                dimnames = list(Hospitalized = c("Yes",
                                                 "No"),
                                Vaccinated = c("Yes",
                                               "No")))

ph <- rowSums(joint)

joint["Yes","Yes"] / ph["Yes"]
```

```
       Yes
0.08536585
```

b) Using Bayes' equation, compute the probability of being hospitalized with covid-19 conditional on being vaccinated, i.e., $Pr(\text{Hospitalized} = \text{Yes}|\text{Vaccinated} = \text{Yes})$. [3 Marks]

$$\begin{aligned}\Pr(\text{Hospitalized=Yes}|\text{Vaccinated=Yes}) &= \frac{\Pr(\text{Vaccinated=Yes}|\text{Hospitalized=Yes})\Pr(\text{Hospitalized=Yes})}{\Pr(\text{Vaccinated=Yes})} \\ &= \frac{0.007}{0.007+0.666} \\ &= \frac{0.007}{0.673} \\ &= 0.010\end{aligned}$$

```
pv <- colSums(joint)

joint["Yes", "Yes"] / pv["Yes"]
```

```
       Yes
0.01040119
```

Question 5

Let $Y_1, ...Y_n$ be i.i.d. draws from the population with mean μ_Y. Type your answers.

a) What is the bias of an estimator? [1 Mark]

Bias of an estimator is the difference between the expected value and the true parameter.

b) Show that $\bar{Y}$ is an unbiased estimator of μ_Y. [5 Marks]

Since $E(Y) = \mu_Y$,

$$\begin{aligned} E(\bar{Y}) &= \frac{1}{n}\sum_{i=1}^{n} E(Y_i) \\ &= \frac{1}{n}(n\mu_Y) \\ &= \mu_Y \end{aligned}$$

Question 6

A researcher claims that the average height of adult men in a city is 175 cm. To test this claim, a random sample of 30 adult men is selected, and their heights are recorded. The sample has a mean height of 172 cm and a standard deviation of 8 cm.

a) State the null and alternative hypotheses for this test. [1 Mark]

$$\begin{aligned} H_0 &: \mu = 175 \\ H_1 &: \mu \neq 175 \end{aligned}$$

b) Calculate the test statistic using R commands. [2 Marks]

$$\begin{aligned} t &= \frac{\bar{Y} - \mu_{Y,0}}{SE(\bar{Y})} \\ &= \frac{172 - 175}{8/\sqrt{30}} \\ &= -2.054 \end{aligned}$$

```
samplemean_act <- 172
mean_h0 <- 175
se <- 8/sqrt(30)

tstatistic <- (samplemean_act - mean_h0) / se

tstatistic
```

```
[1] -2.05396
```

c) Determine the critical value(s) at the 5% significance level using the `qnorm()` command. [1 Mark]

The probability of each side is $0.05/2 = 0.025$, the critical values are 1.96 and -1.96.

```
qnorm(0.025)
```

```
[1] -1.959964
```

```
qnorm(0.975)
```

```
[1] 1.959964
```

d) Based on the test statistic and critical value(s), decide whether to reject the null hypothesis.

Since $t = -2.054 < -1.96$, so we reject H_0.

e) Construct a 95% confidence interval for the true mean height using R commands. [3 Marks]

$$\begin{aligned}\bar{Y} \pm 1.96\, SE(\bar{Y}) &= 172 \pm 1.96 \left(\frac{8}{\sqrt{30}}\right) \\ &= 172 \pm 1.96(1.4606) \\ &= 169.14 \text{ and } 174.86 \text{ cm}\end{aligned}$$

```
172 + c(-1, 1) * qnorm(0.975) * se
```

```
[1] 169.1373 174.8627
```

Question 7

In a randomized controlled trial, 100 participants are randomly assigned to either a treatment group or a control group. The treatment group has an average improvement score of 80 with a standard deviation of 10, while the control group has an average improvement score of 75 with a standard deviation of 12.

a) Formulate the null and alternative hypotheses to test if the treatment has a significant causal effect. [2 Marks]

Let μ_T be the treatment group, μ_C be the control group.

$$H_0 : \mu_T - \mu_C = 0$$

$$H_1 : \mu_T - \mu_C \neq 0$$

b) Compute the test statistic for the difference between means using R commands. [2 Marks]

$$\begin{aligned} t &= \frac{(80 - 75) - 0}{\sqrt{10^2/50 + 12^2/50}} \\ &= 2.263 \end{aligned}$$

```
tstatistic <- (80 - 75) / sqrt(10^2 / 50 + 12^2 / 50)
tstatistic
```

```
[1] 2.263394
```

c) Assuming a significance level of 5%, conclude whether the treatment has a significant causal effect. [2 Marks]

Since $2.263 > 1.96$, so we reject H_0. Therefore the treatment has a positive causal effect on the improvement score.